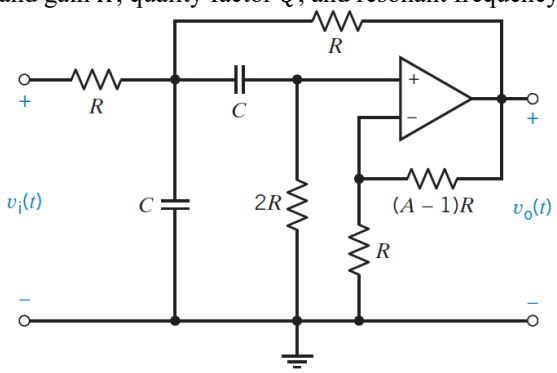
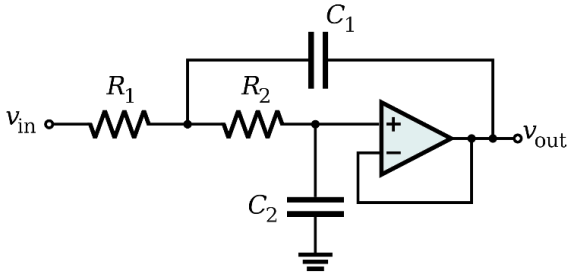
EEL 3112 – Circuits 2 – Fall 2026

HW #8

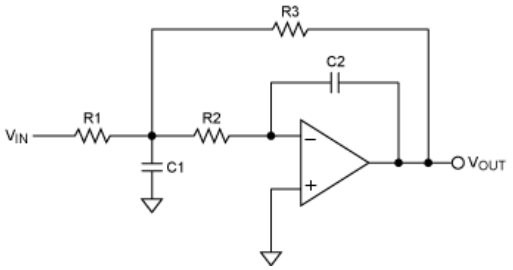
1. Design and compare 4th-order Butterworth and Bessel LPFs with $f_c = \frac{\omega_c}{2\pi} = 100\text{Hz}$ and a gain of 10using your coding language of choice. Include your code.
   a. What are the transfer functions?
   b. Plot the Bode plots for both magnitude and phase of the frequency response functions. For your sketch, please have vertical axis go from $-120\text{dB}$to 40dB horizontal axis go from 1to 10^5.
   c. Plot the group delays
   d. Plot the step function responses
2. Design a 5th -order Butterworth BPF with $f_{c1} = \frac{\omega_{c1}}{2\pi} = 250\text{kHz}$, $\beta = 1.5\text{MHz}$ and a gain of 1. Include your MATLAB code.
   a. What is the transfer function?
   b. Plot the Bode plots for both magnitude and phase of the frequency response function.
   c. What is f_{c2}?
   d. What is the center frequency f_0?
3. Consider the Sallen-Key circuit below.
   a. Upon inspection (by taking low and high frequency limits), what type of filter is this?
   b. Determine the transfer function $H(s) = \frac{V_o(s)}{V_i(s)}$.
   c. What are the flatband gain K, quality factor Q, and resonant frequency ω_0 of the filter?



4. Single ended piezoresistive accelerometers can be modelled as a thevenin voltage source with series resistance. You need to design a 2nd order anti-aliasing lowpass filter to buffer various accelerometers outputs (with different resistances). In your research, you came upon the Sallen-Key and Multiple-Feedback lowpass filter topologies, and need to pick between them.



Sallen-Key Lowpass



Multiple-Feedback Lowpass

a. Find the transfer function H(s)=Vout/Vin for both circuits. Identify w_0, Q, and DC Gain.
b. For your antialiasing filter, you wish to design the filter to have the following design characteristics. Butterworth response (Q=.707), Cutoff frequecy of 2kHz, Unity DC gain, all resistors of equivilant value 10kΩ. Find the capacitor values C1 and C2 for both circuits. You may use the coding lanauage of your choice, an online circuit calculator, spice, or any other method you wish just outline your steps.
c. Assume you have 3 accelerometers of series resistance 1k, 10k, and 100k this circuit needs to work with.You now need to choose between the two circuits. Make a table outlining the DC Gain, corner frequency, and Q for both circuit topologoies.
d. Analyze your results in the table. Which circuit would you choose depending on if you value frequency response or gain. What could you do to the component values or topology to design a single circuit that has stable magnitude and frequency response regardless of drive impedance.

5. Given:
   - Suppose you wish to measure the frequency response function of a shaker-excited structure. The output signal is from an accelerometer. The input signal is the excitation signal (to the shaker) and is band-limited white noise.
   - The frequency range of interest is from 0 to 2 kHz.
   - You are using a 12-bit A/D to acquire your data.
   - You have at your disposal 4th-order low-pass Butterworth filters characterized by a response of –3 dB at the cutoff frequency f_c and a roll-off of 24 dB/octave, and a stopband attenuation of -80 dB due to the noise floor of the electronics.
   - Note: the signal paths are as follows:
     - Voltage input to shaker is split and sent through LPF and then into A/D, the final voltage that is input to the A/D is V_i
     - Voltage output from accelerometer is sent through LPF and then into A/D, the final voltage that is input to the A/D is V_o
     - You are interested in ultimately estimating $H(s) = \frac{V_o(s)}{V_i(s)}$

   Find:

   a) What is the dynamic range of the A/D in dB?
   b) Do the filters need to have linear phase? Why or why not?
   c) Sketch the frequency response magnitude function of the filters. You may assume that, for the same f_c, the two filters will be identical.
   d) Select an appropriate minimum sampling frequency to prevent aliasing. Your goal is to maximize the dynamic range of the A/D.

Hints / Background information:

- A/D converters are typically 8-bit, 12-bit, 14-bit, 16-bit, or 24-bit. The number of bits N represents how many ones and zeroes (bits) are available for the digital output of the A/D converter. For example, an 8-bit converter has 8 bits available.
    - For simplicity, consider a 2-bit A/D converter (N = 2) with a **range** of -5 to 5 volts. The voltage range is divided into **bins** as follows:

Analog voltage (volts)	**Bin number**	**Digital output** (binary number)	**Assigned voltage, V** (volts)
$-5 \leq V < -2.5$	0	00	-3.75
$-2.5 \leq V < 0$	1	01	-1.25
$0 \leq V < 2.5$	2	10	1.25
$2.5 \leq V < 5$	3	11	3.75

- The assigned voltage for each bin is here defined as half-way between the limits of the bin.
- The number of bins = 2^N for an N-bit converter. For example, in the above table, for a 2-bit converter, there are $2^2 = 4$ bins.
    - The voltage **resolution** of the A/D converter is defined
        - $\frac{(V_{\max} - V_{\min})}{2^N}$
        - Applying this to the above example yields $\frac{(V_{\max} - V_{\min})}{2^N} = \frac{(5-(-5))}{2^2} V = \frac{10}{4} V = 2.50V$
    - This resolution is too large for most practical applications! Notice that one cannot tell the difference between an input of 2.6 V and 4.9 V - both of these input voltages will fall in bin number 3, and would be assigned the output voltage of 3.75 V.